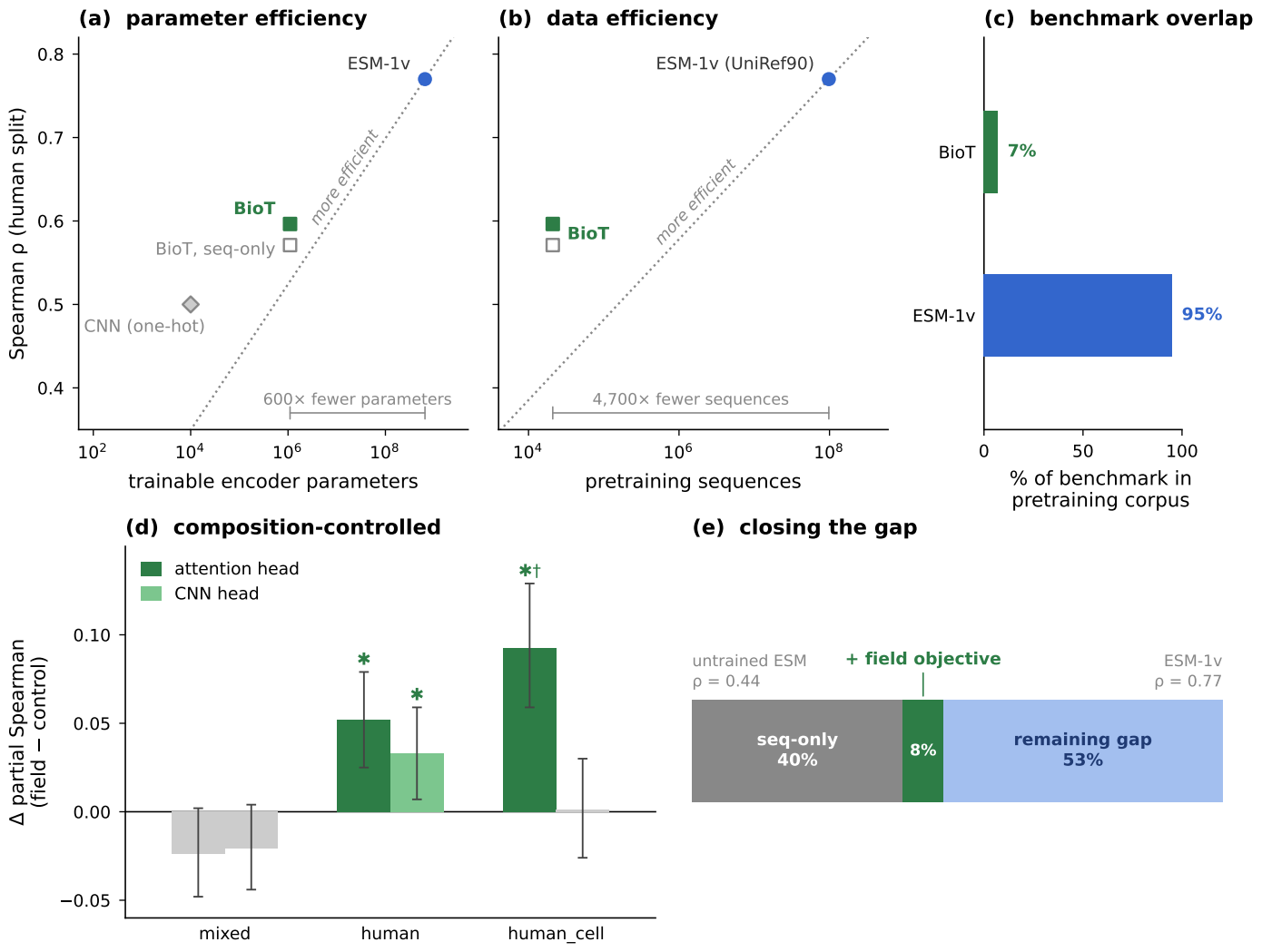
(a) parameter efficiency
Spearman ρ (human split)
0.8
0.7
0.6
0.5
0.4
ESM-1v
more efficient
BioT
BioT, seq-only
CNN (one-hot)
600× fewer parameters
10^2
10^4
10^6
10^8
trainable encoder parameters
(b) data efficiency
ESM-1v (UniRef90)
more efficient
BioT
4,700× fewer sequences
10^4
10^6
10^8
pretraining sequences
(c) benchmark overlap
BioT
7%
ESM-1v
95%
0
50
100
% of benchmark in pretraining corpus
(d) composition-controlled
attention head
CNN head
Δ partial Spearman (field − control)
0.10
0.05
0.00
−0.05
*
*
*†
mixed
human
human_cell
(e) closing the gap
untrained ESM
ρ = 0.44
+ field objective
ESM-1v
ρ = 0.77
seq-only 40%
8%
remaining gap 53%